MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 37 | Date: 12-03-2026

Total Questions: 11 | Answer all questions.

Section 1: MCQ (6 Questions - 12 Marks)

Q1. [2 marks] (Medium)

Here are 10 unique exam questions for your Data Structures (CS101) Mid-Term Examination:

Q2. [2 marks] (Easy)

Which of the following is a non-linear data structure?

Options:
A) Array B) Linked List C) Stack D) Tree

Q4. [2 marks] (Easy)

An Abstract Data Type (ADT) is characterized by:

Options:
A) Its implementation details B) Its specific memory allocation C) Its behavior from the user's perspective D) The programming language used

Q6. [2 marks] (Medium)

Which operation is used to add an element to the top of a stack?

Options:
A) Dequeue B) Pop C) Push D) Inject

Q8. [2 marks] (Medium)

In a queue, the element that was added first is the first one to be removed. This principle is known as:

Options:
A) LIFO B) FIFO C) LILO D) FILO

Q10. [2 marks] (Hard)

Which of the following data structures typically requires extra space proportional to the number of elements (O(N)) for its auxiliary data (e.g., pointers) in addition to storing the actual data elements?

Options:
A) Array B) Singly Linked List C) Contiguous Stack (array-based) D) Static Queue (array-based)

Section 2: Short Answer (5 Questions - 25 Marks)

Q3. [5 marks] (Medium)

Explain the concept of Big O notation in the context of algorithm analysis.

Q5. [5 marks] (Hard)

Compare and contrast arrays and singly linked lists based on memory allocation and insertion/deletion operations in the middle of the structure.

Q7. [5 marks] (Easy)

Briefly explain the purpose of a null pointer.

Q9. [5 marks] (Medium)

Describe one significant advantage of a doubly linked list over a singly linked list.

Q11. [5 marks] (Hard)

Explain a practical scenario where a circular linked list would be a more suitable data structure than a linear (singly or doubly) linked list.

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